

**M.S. RAMAIAH INSTITUTE OF TECHNOLOGY
(AUTONOMOUS INSTITUTE, AFFILIATED TO VTU)
BANGALORE-560054**

SEMESTER END EXAMINATIONS – JUNE/JULY 2024

Course: B.E.

Subject: Computer Aided Engineering Drawing

Subject code: MELC28

Semester: II

Max. marks: 50

Duration: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.
- Draw Solutions using pencil and scale to the dimensions.
- Answer Unit-1 questions in sketch work only, no computer solution is needed
- For Unit-2 & Unit-3, submit both sketch work and computer solutions

Q.No.	UNIT – I	Marks
1.	An equilateral triangular lamina of 50mm side is resting on one of its edge such that the surface of the lamina is inclined at 30° to HP. The edge on which the lamina rests is inclined at 45° to VP.	10
2.	A Square prism of base side 30mm and axis 60mm long rests on HP on one of the square faces such that two adjacent sides of the square faces are equally inclined to VP. Draw the development of the lateral surface of the prism.	10
UNIT – II		
3.	A Hexagonal prism of 25mm sides of base and 50mm axis length rests on HP on one of its edges. Draw the projections of the prism when the axis is inclined to HP at 45 degree and appears to be inclined to VP at 40 degree	20
4.	A Pentagonal prism 25mm sides of base and 60mm axis length rests on HP on one of its edges of the base. Draw the projection of the prism when the axis is inclined to HP at 45° and appears to be inclined to VP at 40°.	20
UNIT – III		
5.	Draw the Isometric projection of Hexagonal Pyramid of Side 30mm & height 70mm placed on Hexagonal prism of side 30mm and height 70mm.	20
6.	A hemisphere diameter 70mm is placed on the ground on its curved surface. A cone base diameter 70mm and height 70mm is placed centrally on it. Draw the isometric projection of combination of solids	20

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Q.No.	UNIT – I	Marks
1.	Draw the projections of a circular plate of negligible thickness of 50mm diameter resting on HP on a point on the circumference, with its plane inclined at 30° to HP and the top view of the diameter passing through the resting point makes 60° with VP.	10
2.	A vertical cylinder of 50mm diameter and 60mm height is cut by a section plane perpendicular to VP and inclined at 30° to the axis so as to pass through the top end of one of the extreme generator in the front view. Draw the development of lateral surface of the truncated cylinder.	10
UNIT – II		
3.	A square prism 30mm size of base and 60mm axis length rests on HP on one of its edges of the base which is inclined at 30° to VP. Draw the projection of the prism when the axis is inclined to HP at 45° .	20
4.	A pentagonal pyramid of side 25mm & axis 60mm rests on HP on one of its edges of the base which is inclined to VP at 40° . Draw the projection of the pyramid when the axis of the pyramid is inclined to HP at 40° .	20
UNIT – III		
5.	A frustum of cone 30mm top face diameter and 60mm base diameter rests vertically on a cylinder block of 70mm diameter and 30mm height such that the axis of both the solids line in the same line. Draw the isometric projection of the combination of the solids.	20
6.	A rectangular pyramid of base 40mmx25mm and height 50mm is placed centrally on a rectangular slab sides 100mmx60mm and thickness 20mm. Draw the isometric projection of the combination.	20

height 40mm rest

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Q.No.	UNIT – I	Marks
1.	A hexagonal lamina of 30mm sides with one of its corners touching VP and surface inclined at 45° to it. One of its edges is appears to be inclined to HP at 30° . Draw the front and top views of the lamina in its final position	10
2.	A vertical cone 40mm diameter of base and height 50mm is cut by a cutting plane perpendicular to VP and inclined at 30° to HP so as to bisect the axis of the cone. Draw the development of lateral surface of the truncated cone.	10
	UNIT – II	
3.	A pentagonal prism 25mm sides of base and 50mm axis length rests on HP on one of its corner of the base such that the two base edges containing the corner on which it rests makes equal inclinations with HP. Draw the projections of the prism when the axis of the prism is inclined to HP at 40° and appears to be inclined to VP at 45° .	20
4.	A pentagonal pyramid of base edge 25mm and height 60mm resting on HP with one of its triangular face. The axis of the pyramid is appearing to be at 35° to VP. Draw projections	20
	UNIT – III	
5.	Draw the isometric projection of the combination of solids formed by a frustum of cone and co-axial frustum of pentagonal pyramid. The lower frustum of cone is of 80mm base diameter, 60mm top diameter and height 25mm. The upper frustum of pyramid is of 30mm side of base, 20mm side of top face and height 40mm.	20
6.	A cylindrical slab 50mm diameter and 20mm thickness is surmounted by a cube of 30mm side. The axes of the solids are in the same line. Draw the isometric projection of combination of solids.	20

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Q.No.		Marks
UNIT – I		
1	An equilateral triangular lamina of sides 25mm rests on one of its sides on VP. The lamina makes 45° to VP and the side on which it rests makes 45° to HP. Draw its projections.	10
2	A hexagonal pyramid of 25 mm sides of base and 60 mm axis length rests on HP. The pyramid is cut at the axis by section plane of 25° to the HP at distance 30 mm from the base. Draw the development of lateral surface of a frustum of the pyramid	10
UNIT – II		
3	A square prism of side 30mm and axis 60mm long rests on one of its shorter edges on HP. Draw the projections of the prism when the axis is inclined to HP at 40° and appears to be inclined to VP at 45°	20
4	A cone of base diameter 40mm and axis 50mm is resting on one of the points of the circumference on HP such that the axis is inclined at 40° to HP and appears to be inclined at 30° to VP. Draw its projections.	20
UNIT – III		
5	A hexagonal prism of side of base 30mm and axis 50mm is having a hemisphere of top diameter 50mm located centrally on its top face. Draw the isometric projection of the combinations when the top diameter of the hemisphere is facing the top face of the pentagonal <u>hexagonal</u> prism.	20
6	A hemisphere of diameter 40mm is placed centrally on top of a frustum of a hexagonal pyramid of base side 40mm, top face side 20mm and height 50mm, Draw the isometric projection of the combination.	20

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UNIT – I		Marks
Q. No.		
1.	A pentagonal plane with edges measuring 30 mm is positioned on the horizontal plane, with one of its corners such that the edge opposite to this corner is 20 mm above the horizontal plane and forms a 45° angle with the vertical plane. Create the projections of this configuration and determine the inclination of the plane with respect to the horizontal plane.	10
2.	A hexagonal pyramid with 25 mm sides at its base and an axis length of 60 mm is positioned on the horizontal plane. The pyramid is intersected along its axis by a section plane making a 30° angle with the horizontal plane at a distance of 30 mm from the base. Develop lateral surface of a frustum of a pentagonal pyramid.	10
UNIT – II		
3.	A square pyramid with base sides measuring 25 mm and an axis length of 50 mm is positioned on the horizontal plane with one of its base edges resting on it. Render the projections of the pyramid with the axis inclined to the horizontal plane at a 45° angle and appears to be inclined to the vertical plane at a 30° angle	20
4.	A pentagonal prism with a base side length of 25 mm and a height of 50 mm is positioned on the horizontal plane with one of its base edges resting on it. Render the projections of the prism such that its axis is inclined at a 45° angle to the horizontal plane and appears to be inclined at a 30° angle to the vertical plane.	20
UNIT – III		
5.	A regular hexagonal prism with a base edge of 30 mm and an axis length of 60 mm is centrally mounted over a cylindrical block with a diameter of 80 mm and a thickness of 30 mm. Draw the isometric projection of the combined solid.	20
6.	cone with a base diameter of 30 mm and a height of 40 mm rests centrally on a cube with a side length of 50 mm. Draw the isometric projection of the combined solid.	20

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Q.No.	UNIT – I	Marks
1.	An isosceles triangular plate has base 50mm long and altitude 70mm. It is so placed that in the front view it is seen as an equilateral triangle of 50mm sides with one of its sides inclined at 45° to the XY line. Draw its top and front views.	10
2.	Draw the lateral surface of a Hexagonal prism with its sides on hp having side 40mm with a height of 60 mm.	10
UNIT – II		
3.	A square prism 35mm sides of base and 60mm axis length rests on HP on one of its edges of the base . Draw the projection of the prism when the axis is inclined to HP at 40° and the top view of the axis is inclined at an angle of 30° to VP.	20
4.	A Pentagonal pyramid 30mm sides of base and 60mm axis length rests on HP on one of its edges of the base. Draw the projections of the pyramid when the axis is inclined to HP at 45° and the top view of the axis is inclined at an angle of 30° to VP	20
UNIT – III		
5.	Draw the isometric projections of the combination of solids. A cone of 40mm diameter and 80mm high is placed vertically on a cylinder block of 60mm diameter and 40mm thick such that both the solids have a common axis.	20
6.	Draw the isometric projections of the combination of solids. On a rectangular prism of 60x80x20mm thick a triangular pyramid of side 40mm with a height of 50 mm is placed such that the axes of the solids have a common axis.	20

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Q No.	UNIT - I	Marks
1	A hexagonal plane with edges measuring 30 mm is positioned on the horizontal plane, with one of its corners such that the edge opposite to this corner is 15 mm above the horizontal plane and forms a 40° angle with the vertical plane. Create the projections of this configuration and determine the inclination of the plane with respect to the horizontal plane.	10
2	A hexagonal prism with 25 mm sides at its base and an axis length of 60 mm is positioned on the horizontal plane. The prism is intersected along its axis by a section plane making a 30° angle with the horizontal plane at a distance of 30 mm from the base. Develop the unfolded lateral surface of a frustum of a hexagonal prism.	10
UNIT - II		
3	A hexagonal pyramid with base sides measuring 25 mm and an axis length of 50 mm is positioned on the horizontal plane with one of its base edges resting on it. Render the projections of the pyramid with the axis inclined to the horizontal plane at a 45° angle and appears to be inclined to the vertical plane at a 30° angle.	20
4	A pentagonal prism with a base side length of 25 mm and a height of 50 mm is positioned on the horizontal plane with one of its base edges resting on it. Render the projections of the prism such that its axis is inclined at a 40° angle to the horizontal plane and appears to be inclined at a 30° angle to the vertical plane.	20
UNIT - III		
5	A regular hexagonal pyramid with a base edge of 30 mm and an axis length of 60 mm is centrally mounted over a cylindrical block with a diameter of 80 mm and a thickness of 30 mm. Draw the isometric projection of the combined solid.	20
6	A cone with a base diameter of 30 mm and a height of 40 mm rests centrally on a cube with a side length of 60 mm. Draw the isometric projection of the combined solid.	20

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- For Unit-2 & Unit-3, submit both sketch work and computer solutions

No.	UNIT – I	Marks
1.	A Pentagonal lamina of sides 30mm rests on one of its corner on VP. The lamina makes 45° to VP and the side opposite the resting corner makes 45° to HP. Draw its projections.	10
2.	A hexagonal pyramid of 25mm side and 50mm height is resting on HP such that one of the base edge is perpendicular to VP. A section plane perpendicular to VP bisect the axis at 30° to HP. Draw the development of lower portion of the pyramid.	10
UNIT – II		
3.	A rectangular base prism of 30mm x 40mm side and 60mm height is resting on HP on one its shorter edges. The axis of the prism is inclined 40° to HP and appears to be inclined 50° to VP. Draw the projections of the prism	20
4.	A Pentagonal pyramid of 25mm side and 60mm height is resting on HP on one its triangular surface. The axis of the pyramid is appears to be inclined 45° to VP and the base is nearer to the observer. Draw its projections.	20
UNIT – III		
5.	A hemisphere of diameter 40mm is placed centrally on top of a frustum of a pentagonal pyramid of base side 40mm, top face side 20mm and height 50mm. The axis of the both the solids are coaxial. Draw the isometric projection of the combination.	20
6.	A square base pyramid of base side 40mm and height 60mm is placed centrally on the top face of the cylindrical block of 60mm diameter and 50mm height. The axis of the both the solids are coaxial. Draw the Isometric projections of the combined solid.	20

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Q.No.	UNIT – I	Marks
1.	A pentagonal lamina of edges 30 mm is resting on HP with one of its corners such that the edge opposite to this corner is 20mm above HP and makes an angle of 40° with VP. Draw the top and front views of the lamina in this position. Determine the inclination of the lamina with HP.	10
2.	A rectangular pyramid of side of base 30 mm X 40 mm and axis 50 mm is resting on its base on HP and one of the shorter base edges is parallel to VP. The pyramid is cut by a section plane which is perpendicular to VP, inclined at 30° to HP and bisecting the axis of the pyramid. Obtain the development of the truncated pyramid.	10
UNIT – II		
3.	A hexagonal prism of side of base 25 mm and axis 50 mm is resting on one of its base edges on HP such that the edge on HP is inclined at 30° to VP. Draw its projections when the axis makes an inclination of 40° with HP.	20
4.	A pentagonal pyramid of base side 25 mm and 50 mm axis length rests on HP on one of its slant faces. Draw the projections of the pyramid when the axis of the top view is inclined to VP at 30° .	20
UNIT – III		
5.	A cone with a base diameter of 40 mm and height of 60 mm rests centrally over a frustum of a hexagonal pyramid with the base sides of 40 mm, top side of 35 mm, and height of 55 mm. Draw the isometric projection of the combined solid.	20
6.	A sphere of diameter 50mm is placed centrally on top of a frustum of a hexagonal pyramid of base edges 40mm, top edges 30mm and height 60mm. Draw the isometric projection of the combination of the solids.	20

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1	An equilateral triangular lamina of sides 25mm rests on one of its sides on VP. The lamina makes 45° to VP and the side on which it rests makes 45° to HP. Draw its projections.	10
2	A hexagonal pyramid of 25 mm sides of base and 60 mm axis length rests on HP. The pyramid is cut at the axis by section plane of 25° to the HP at distance 30 mm from the base. Draw the development of lateral surface of a frustum of the pyramid	10
UNIT – II		
3	A square prism of side 30mm and axis 60mm long rests on one of its shorter edges on HP. Draw the projections of the prism when the axis is inclined to HP at 40° and appears to be inclined to VP at 45°	20
4	A cone of base diameter 40mm and axis 50mm is resting on one of the points of the circumference on HP such that the axis is inclined at 40° to HP and appears to be inclined at 30° to VP. Draw its projections.	20
UNIT – III		
5	A hexagonal prism of side of base 30mm and axis 50mm is having a hemisphere of top diameter 50mm located centrally on its top face. Draw the isometric projection of the combinations when the top diameter of the hemisphere is facing the top face of the pentagonal <i>hexagonal</i> prism.	20
6	A hemisphere of diameter 40mm is placed centrally on top of a frustum of a hexagonal pyramid of base side 40mm, top face side 20mm and height 50mm, Draw the isometric projection of the combination.	20